



3D-printed sculptures (Credits: www.sculpteo.com and <https://ultimaker.com>)

the printer. To reach the desired finish, printed objects are further processed. Choice of material depends on printer type, which ranges from plastics and rubber to sandstone, metals and alloys.

Software for 3D printing

Selection of software for your prints is based on your application modelling, sculpting or customising requirements. There is a diverse range of software available for beginners, intermediate designers and professionals. There are software that are freely available (open source) or that can cost you thousands per licence per year. These software can also be used for designing furniture or fabrics, aerospace or transportation, and fashion, among others.

Software such as SketchUp, Autodesk 123D, Ultimaker Cura and Free CAD are useful for beginners and intermediate designers. Whereas, for intermediate designers and professional users, Blender, 3ds Max, AutoCAD, Autodesk Maya and Fusion 360 are recommended.

3ds Max focuses on animation and offers some great 3D modelling features such as shading tools, polygon modelling and parametric mesh modelling, among others.

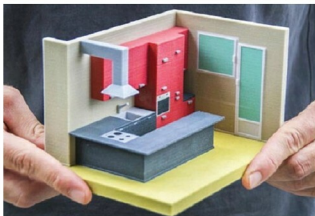
Cura can be used with almost any 3D printer. 3D models of any object can be created by taking photos of objects. This software is also available for smartphones. It is ideal for collaborative efforts as it allows

you to make changes, leave comments and put up red lines. For rapid prototyping, you can use it in basic mode, but for more precise control, you need to switch to expert mode. Cura can also be used for direct control over the machine connected to the PC for said duration.

SketchUp is an open source software, which provides all basic features when you download the free STL export module and install it. Its professional edition is available for architects, engineers and interior designers.

Fusion 360 provides a cloud platform that enables you to store the entire history of the model including the changes made to it. Various design options are available for students and educators, as you can apply for a three-year educational licence.

Blender has a large number of tutorials freely available. This open source software has a steep learning curve, and is ideal for users who feel ready to design complex 3D models.



3D-printed model of a modular kitchen using full-coloured sandstone (Credit: www.3dhubs.com)

Maya provides smooth surfaces and shapes. It is great for artistic printing requirements. It can be used for highly-detailed models because of its high rendering engine. It allows colouring with smooth operation and realistic representations.

FreeCAD is an open source 3D modelling tool, which enables you to design real-life objects of any size. It lets you change parameters from previous models and tweak these as per your desire. It is a great training tool, but it is not appropriate for professionals.

Software Blender, Slic3R and Cura for 3D modelling and printing are provided in the DVD that comes with this month's issue of *Electronics For You Plus*.

Challenges

There are a few challenges with 3D printers like limited materials availability, lack of expertise and/or training among workforce/employees, longer production timelines, limited recyclability, data storage requirements and so on.

Before selecting a 3D printer, the following questions need to be answered to be able to select the best 3D printer for your needs:

1. What do you want to print?
2. What size objects do you want to print?
3. What materials do you want to print with?
4. Do you want to print in multiple colours?
5. What surface should you build on?
6. Do you need a closed frame?
7. How do you want to connect to the printer?
8. What software do you need?

After a comparison study based on weight, material supported, wireless connectivity, resolution, technology and LCD screen, I found five great 3D printers, namely, Makerbot Replicator+, Ultimaker 3, Formlabs, Raise3D and XYZ. But in the end, you need to decide for yourself which printer suits your application and pocket. **EFY**